Pathway and Network Analysis of Neoadjuvant Chemotherapy Nonresponders

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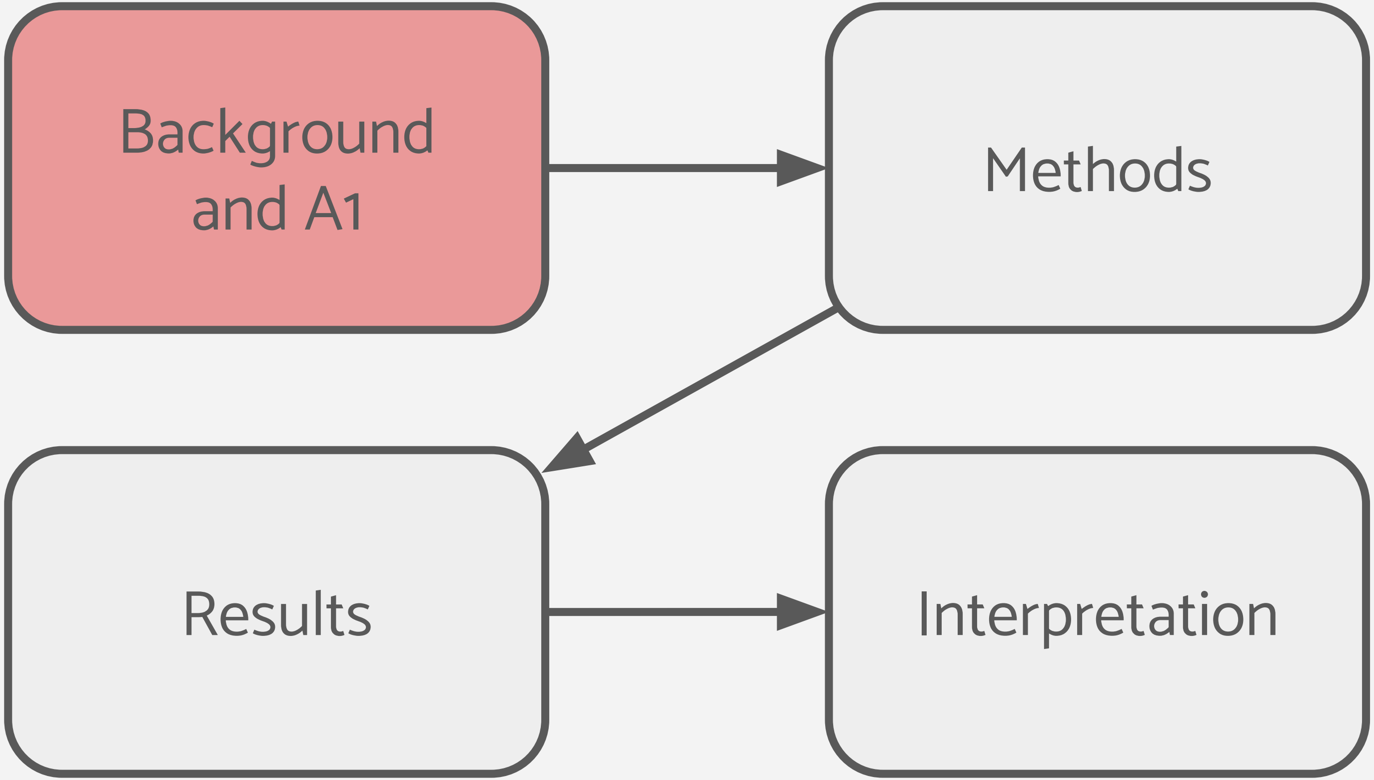
Presentation Roadmap



Background

What changes occur in breast cancer tumors that fail to respond to neoadjuvant chemotherapy (NAC)?

Experimental Design (Guevara-Nieto et al., 2025)

- Breast cancer patients
- Paired (before and after chemotherapy)
- Chemotherapy *nonresponders*

Dataset

- GEO accession GSE309004
- 27 patients, paired samples
- 19,736 genes x 54 samples

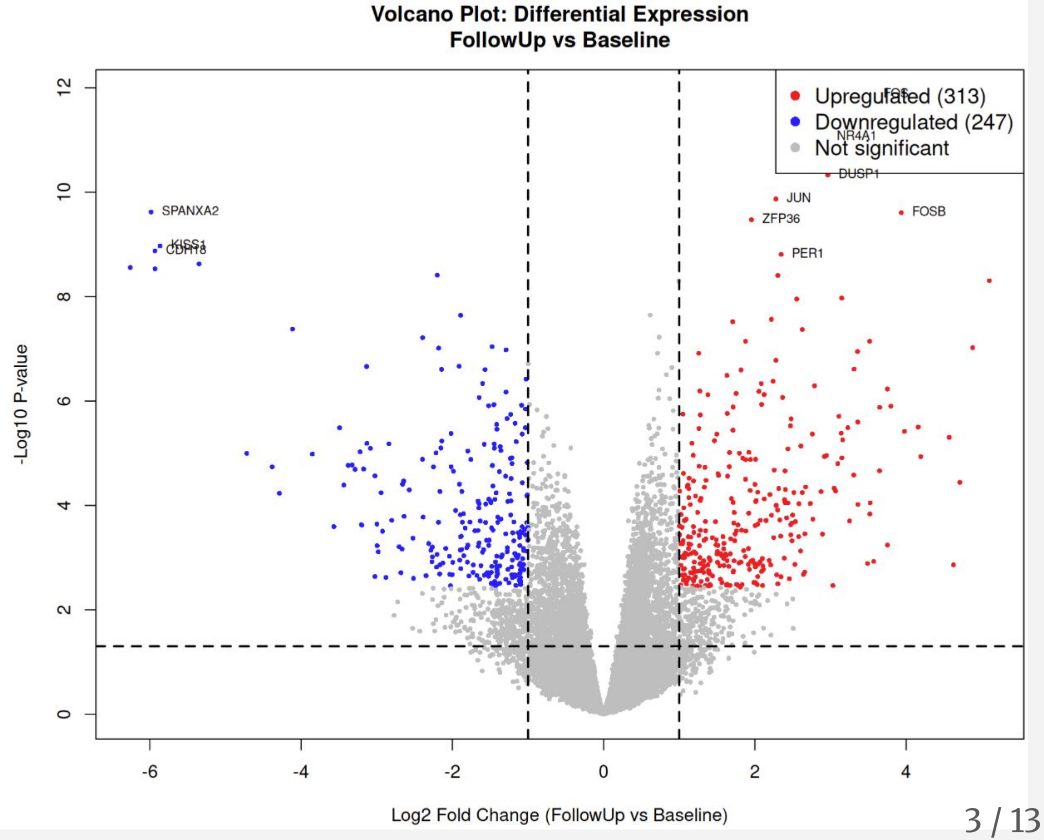
Normalization and Scoring

Normalization

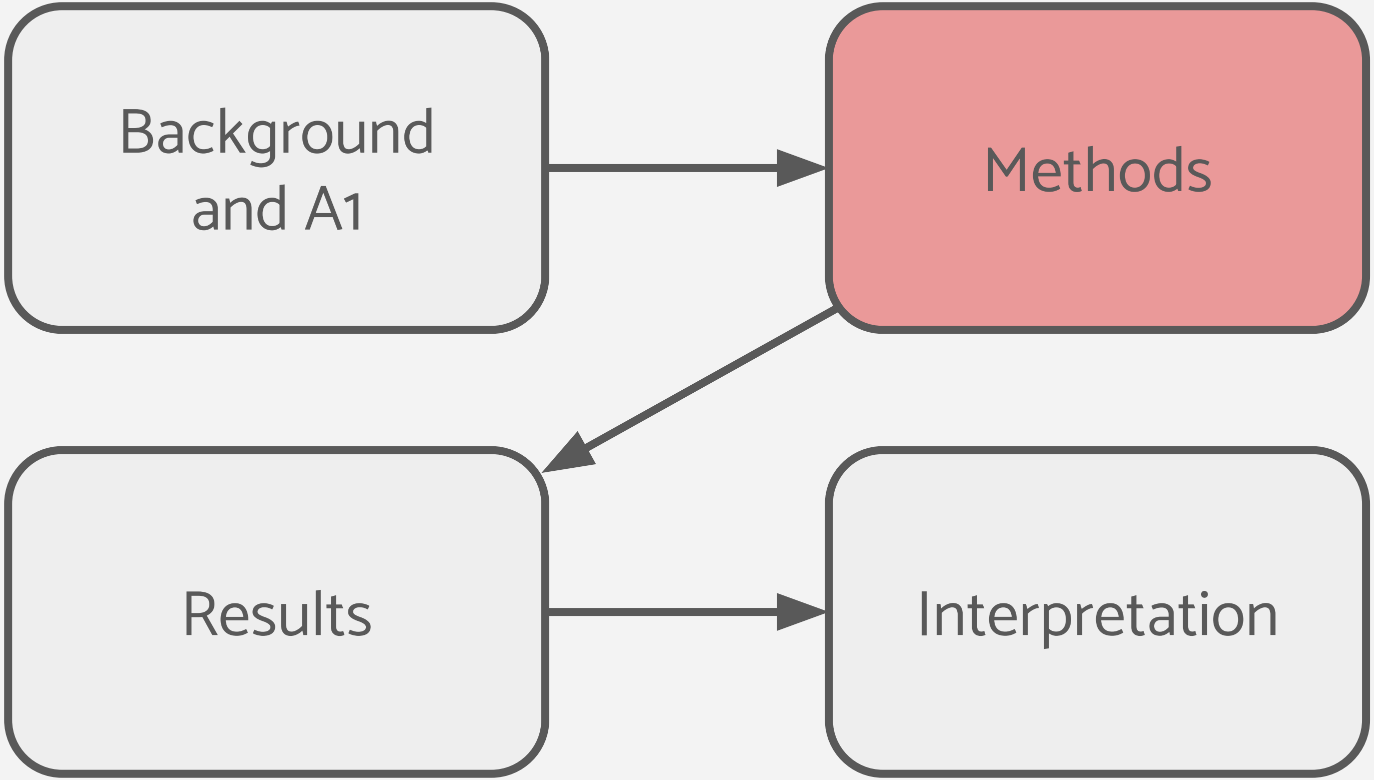
- Authors already normalized
  - Median-of-ratios (MoR)
- I chose TMM normalization
  - Better for noisy data than MoR
  - Produced lower coefficient of variation

Scoring

- Quasi-likelihood generalized linear model
  - No info on true noise -> quasi-likelihood
- Significance filtering
  - logFC and FDR-corrected p-value
  - FDR < 0.05, and |log2 fold change| > 1



Presentation Roadmap



Thresholded ORA

- Used **clusterProfiler** (Yu et al., 2012; Wu et al., 2021) with GO Biological Process database (Ashburner et al., 2000; org.Hs.eg.db, Carlson 2024)
    - clusterProfiler: well-documented, uses BH correction
    - GO:BP: broad and granular

ORA thresholds: Adjusted p-value < 0.05, min gene set size = 10, max = 500 (Subramanian et al., 2005)

```r
# ORA
ora_up <- enrichGO(gene = as.character(up_entrez),
                   OrgDb = org.Hs.eg.db, ont = "BP",
                   pAdjustMethod = "BH", universe = as.character(bg_entrez))
```

Non-Thresholded GSEA

Used **fgsea** (Korotkevich et al., 2021) with MSigDB Hallmark collection (v2023.2.Hs) gene sets

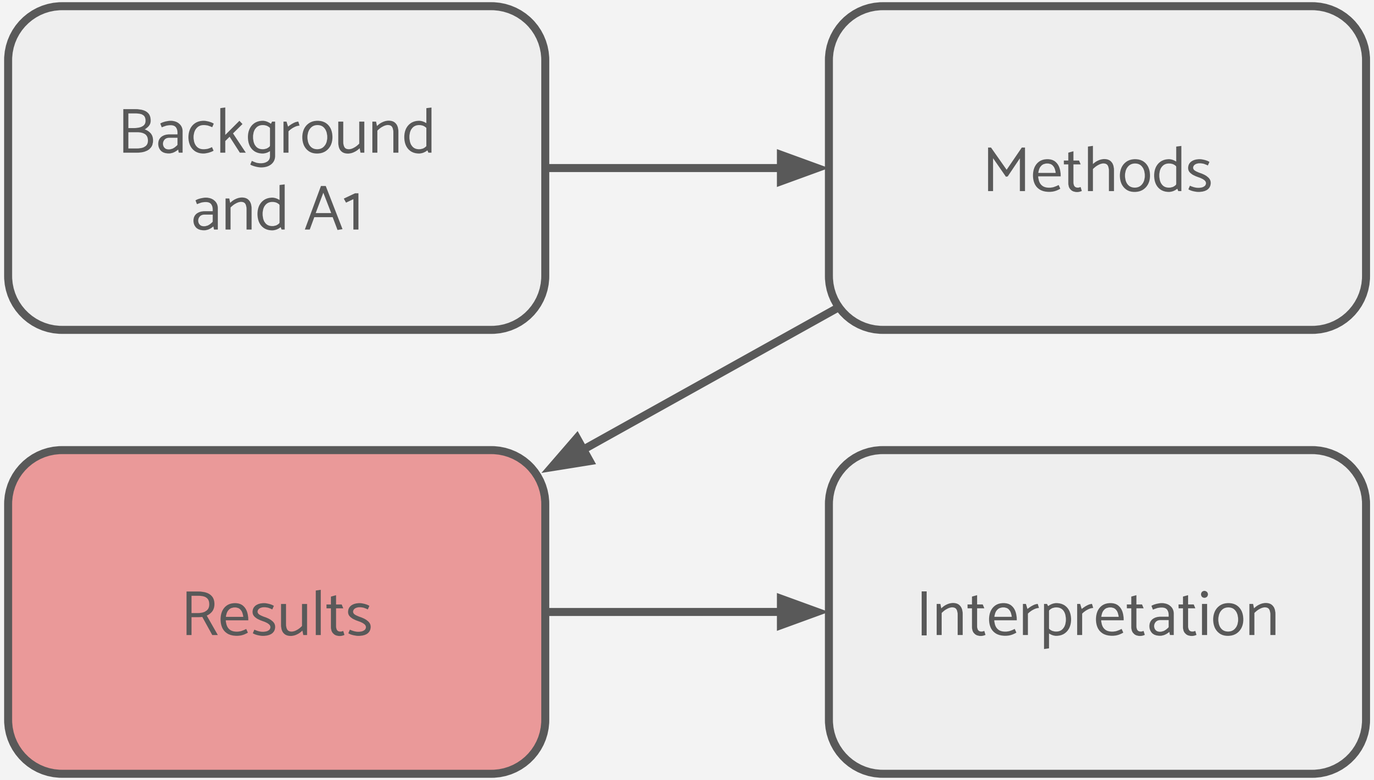
- Fgsea: computationally efficient
- Hallmark gene sets: very well annotated, non-redundant, and clean, easy to view

GSEA parameters: minSize = 15, maxSize = 500, nperm = 1000; significance threshold FDR < 0.25 (Subramanian et al., 2005)

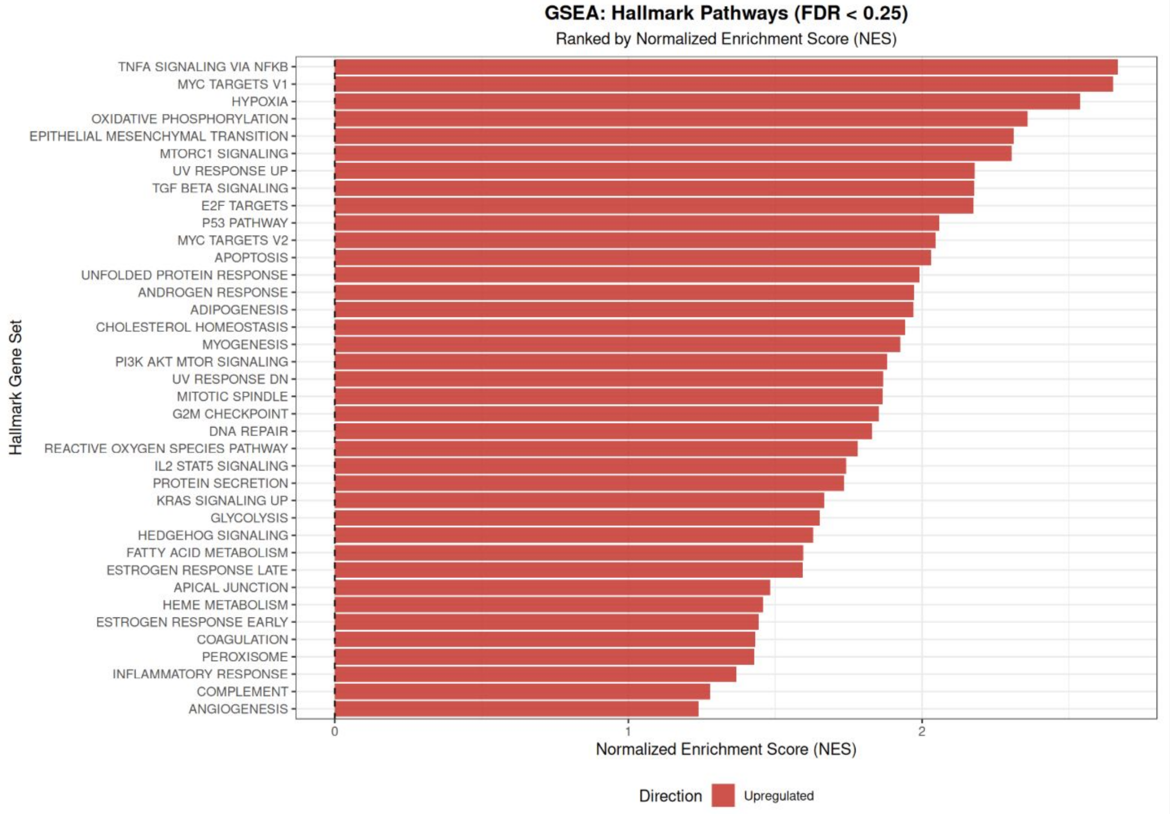
```r
# GSEA ranking metric
results_table$rank_metric <- -log10(results_table$PValue) * sign(results_table$logFC)

# fgsea
fgsea_results <- fgsea(pathways = hallmark_list, stats = ranked_genes,
                       minSize = 15, maxSize = 500, nperm = 1000)
```

Presentation Roadmap



Enrichment Results



ORA (upregulated genes): Stress response and AP-1/MAPK signaling

- Top genes: FOS, JUN, FOSB, ATF3, NR4A1

ORA (downregulated genes): Cell adhesion and tissue organization

- Top genes: CDH18, KISS1, SPANXA2

GSEA: 38 significantly enriched pathways, **all in FollowUp**

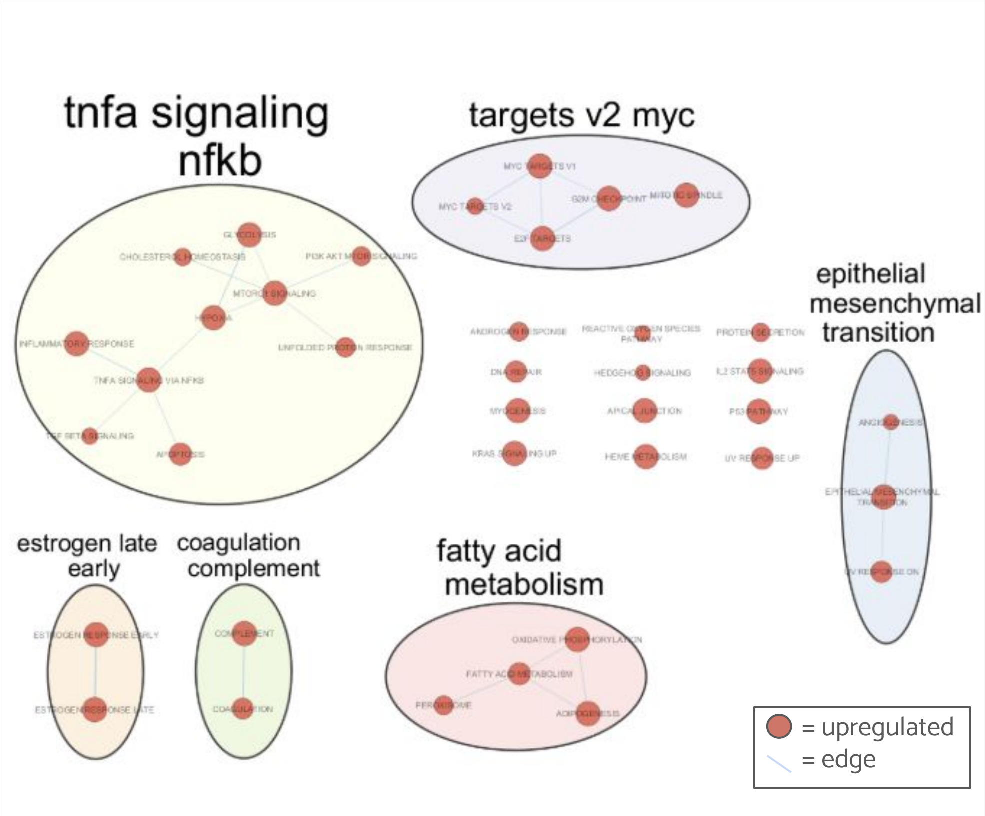
Enrichment Map

Enrichment map network from GSEA results

- **38 nodes** (pathways) and **23 edges** (all nodes enriched in the same direction (post-NAC)
- Edge similarity: Jaccard+Overlap Combined, cutoff = 0.1

AutoAnnotate

- **6 clusters** and **12 singletons**



Major Themes

TNFA Signaling via NF-κB: Major inflammatory pathway

Epithelial-Mesenchymal Transition (EMT): Process by which tumour cells lose cell adhesion

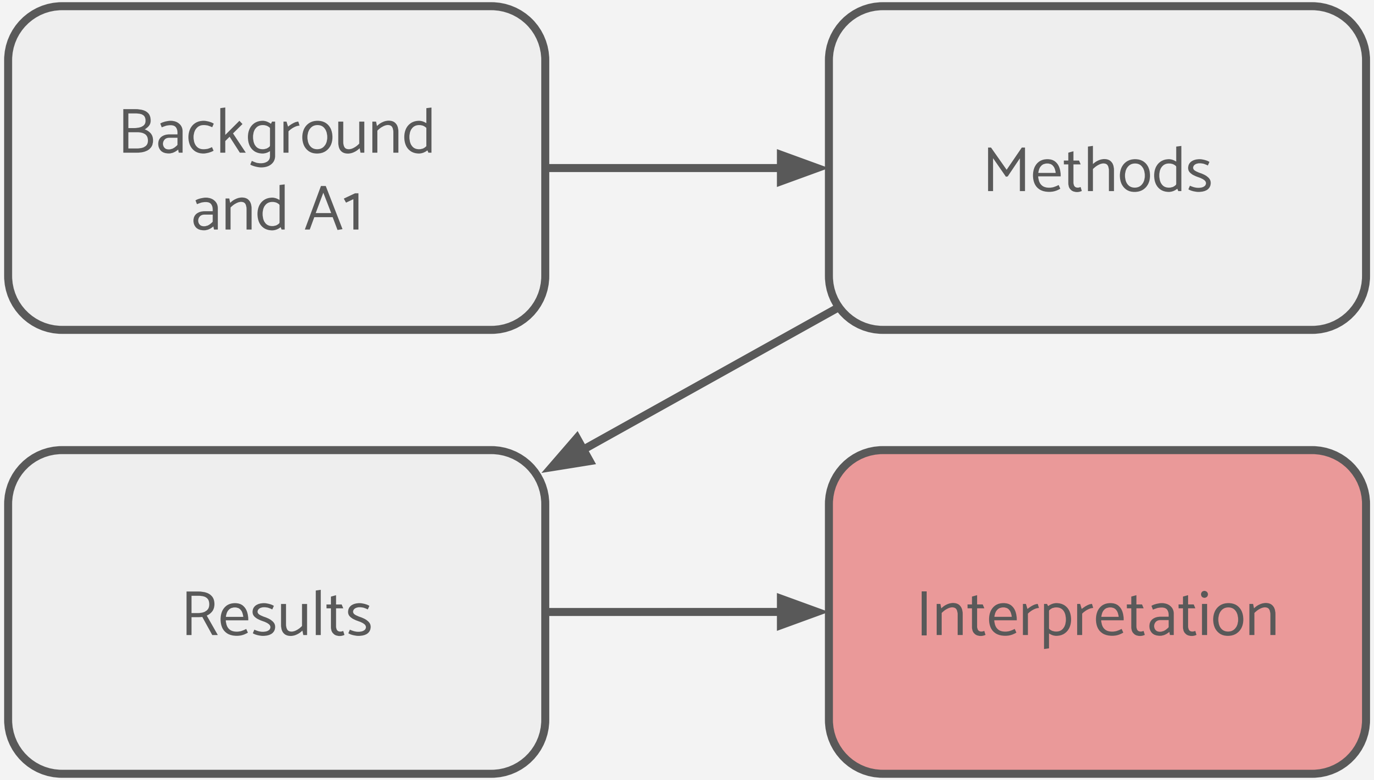
MYC Targets V2: MYC is a transcription factor that drives cell proliferation

Fatty Acid Metabolism*: Reprogramming of how cells can generate energy*

Coagulation/Complement: Immune activation and recruitment

Estrogen Response (Early/Late): Estrogen-driven gene expression

Presentation Roadmap



ORA vs GSEA Comparison

Design differences:

- Different inputs (binary gene list vs continuous ranking) and databases (GO:BP vs Hallmark)
- **Direct term-by-term comparison is not meaningful**

Shared:

- **Inflammatory/stress response** post-NAC
- **Cell adhesion decrease** post-NAC

GSEA only:

- **MYC Targets**
- **Fatty Acid Metabolism**

These signals are likely consistent across many genes but too moderate for individual genes to clear ORA's threshold

Interpretation and Insights

Fatty Acid Metabolism:

- ORA found this cluster enriched post-NAC
- Suggests surviving tumor cells switched from glucose to fatty acid oxidation as an alternative/ additional fuel source
- Lv et al. (2022) showed that chemotherapy-resistant cells upregulate fatty acid oxidation as a key metabolic adaptation to survive drug-induced oxidative stress

Cell adhesion changes:

- ORA: cell adhesion genes (CDH18, KISS1) significantly downregulated, GSEA found the EMT Hallmark cluster enriched post-NAC
- Suggests surviving tumor cells are becoming more migratory
- Galluzzi et al., 2017 discussed this as a strategy for malignant cells

References

R Packages

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Questions?